

SADP

Course: BTECH (CS01)

Set: B | Time: 3 Hours | Max Marks: 35 | Date: 15-03-2026

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

GENERAL INSTRUCTIONS:
Write all the questions

SECTION A: MCQs

1. Which architectural characteristic is a defining feature of Reduced Instruction Set Computer (RISC) processors? [2 Marks]

- a) Variable-length instruction sets
- b) Extensive use of complex addressing modes
- c) Fixed-length instructions and single-cycle execution
- d) Direct execution of high-level language statements

2. In Pulse Width Modulation (PWM), what specific parameter is adjusted to control the average power delivered to a load? [2 Marks]

- a) Signal frequency
- b) Duty cycle
- c) Phase shift
- d) Signal amplitude

3. Which communication protocol is characterized as synchronous, full-duplex, and uses a Master-Slave configuration with a Chip Select (CS) line? [2 Marks]

- a) UART
- b) CAN
- c) USB
- d) SPI

4. Which scheduling policy in a Real-Time Operating System (RTOS) assigns the highest priority to the task with the shortest period? [2 Marks]

- a) Earliest Deadline First (EDF)
- b) Rate Monotonic Scheduling (RMS)
- c) First-Come, First-Served (FCFS)

d) Round Robin (RR)

5. What is the primary purpose of using a "re-entrant" function in a multi-tasking embedded environment? **[2 Marks]**
- a) To ensure the function can be interrupted and safely called again by another task
 - b) To reduce the memory footprint of the code
 - c) To speed up the execution of the function using hardware acceleration
 - d) To allow the function to bypass the kernel scheduler

SECTION B: SHORT ANSWERS

6. Explain the fundamental role of an Analog-to-Digital Converter (ADC) in an embedded system. **[5 Marks]**
7. Describe the function of the Controller Area Network (CAN) bus and state one reason for its popularity in automotive electronics. **[5 Marks]**
8. Distinguish between a Mutex and a Binary Semaphore in the context of task synchronization within an RTOS. **[5 Marks]**
9. Identify two common power-saving techniques implemented in modern embedded systems to extend battery life. **[5 Marks]**
10. Outline the sequence of operations performed by an RTOS kernel during a Context Switch. **[5 Marks]**

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